DATABASES AND SQL

Simple, beginner-friendly study notes

These notes explain the main ideas shown in the lecture slides: databases, why we use them, SQL and NoSQL, SQL language, and tables in SQL.

1. What Is a Database?

A database is an organized collection of data. It stores information in a way that makes the information easy to find, access, manage, and update.

Examples of data stored in a database include:

- Student names, IDs, and marks
- Customer names, phone numbers, and addresses
- Product names, prices, and stock information
- Usernames, email addresses, and social-media activity

2. Why Do We Use Databases?

Databases are useful for several reasons:

- **Store large amounts of data:** A database can keep a large amount of information in an organized place.
- **Security:** Access can be controlled so that only authorized people can view or change certain data.
- **Scalability:** A database system can be expanded to handle more data or more users as needs grow.
- **Easy data changes:** It is easier to insert new data, update existing data, and delete unwanted data.
- **Quick access:** Data can be searched and retrieved more easily than from many separate paper records or files.

3. SQL vs NoSQL

SQL and NoSQL are two broad types of database systems. They organize and store data in different ways.

Feature	SQL Database	NoSQL Database
Meaning	Relational database	Non-relational database
How data is stored	Usually in tables with rows and columns	May use documents, key-value pairs, graphs, or wide-column structures
Data structure	Usually follows a defined structure (schema)	Often allows a more flexible structure
Examples	MySQL, Oracle, PostgreSQL	MongoDB, Cassandra, Neo4j
Common use	Structured data and relationships between tables	Flexible data models or applications that need particular scaling and data-storage patterns

Remember: SQL databases are generally table-based and relational. NoSQL databases are non-relational and can store data in several different formats. NoSQL does not mean that a database cannot use any query language; it means that it does not follow the traditional relational model.

4. What Is SQL?

SQL stands for **Structured Query Language**. It is a language used to communicate with relational databases. We use SQL to create tables and to insert, read, update, or delete data.

Some common SQL commands are:

Command	Purpose
CREATE	Creates a database object, such as a table.
INSERT	Adds new records to a table.
SELECT	Reads or retrieves data from a table.
UPDATE	Changes existing data.
DELETE	Removes records from a table.

5. What Is a Table in SQL?

In a relational database, data is commonly stored in tables. A table organizes information into rows and columns. Each table usually stores information about one type of thing, such as users, students, or products.

5.1 Columns (Fields)

A column is a vertical part of a table. It describes one type of information. For example, a User table may have columns named Id, Name, Email, Followers, and Following.

5.2 Rows (Records)

A row is a horizontal part of a table. It contains all the information about one item or one person. Each row in a User table can represent one user.

6. Example: A User Table

The following example shows a table with five columns and four rows of user data:

Id	Name	Email	Followers	Following
1	Adam	adam@yahoo.in	123	145
2	Bob	bob123@gmail.com	200	200
3	Casey	casey@email.com	300	306
4	Donald	donald@gmail.com	200	105

- **Id:** The unique identification number of a user.
- **Name:** The user's name.
- **Email:** The user's email address.
- **Followers:** The number of people who follow the user.
- **Following:** The number of people the user follows.

In this example, there are 5 columns and 4 data rows. The first row at the top contains the column names (the headings). Adam's information is one record, Bob's information is another record, and so on.

7. Quick Revision

- Database: An organized collection of data.
- SQL: A language used to communicate with relational databases.
- SQL database: Stores data mainly in related tables.
- NoSQL database: Stores data using formats such as documents, key-value pairs, or graphs.
- Table: Organizes data into rows and columns.
- Column: A vertical field that describes one type of information.
- Row / Record: A horizontal set of values describing one item or person.
- Schema: The planned structure of a database or table, including its columns and data types.